CIRC HQ STUDENT HANDOUT | GRADE 5

Helping Hand

Goal: Build a hand-operated tool that safely grips and moves an object beyond normal reach.

35 minutes | 8 steps | Engineering and Design | Playbook A meeting 16, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Demonstrate one successful move and explain how the design serves the stated need.

You will use

- Cardboard
- Craft sticks
- String
- Rubber bands
- Brass fasteners
- Foam pads
- Lightweight test objects
- Scissors
- Role cards
- Build-lead marker
- Test stand

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Read one role card.
- Join the one shared educational prototype team.

2 Step 2: Build safely — 4 min

- Cut only on the table.
- Keep fingers outside the cutting path.
- Carry scissors closed.
- Keep the prototype away from bodies.

3 Step 3: Plan the grip — 5 min

- Choose one lightweight test object.
- Sketch the handle.
- Sketch the moving connection.
- Mark the soft part that holds the object.

4 Step 4: Build one shared part — 8 min

- Do only the work on your role card.
- Put your finished work in the team tray.
- Show it to the build lead.

5 Step 5: Test on the stand — 6 min

- The test team places the build on the test stand.
- The test team moves one approved object.
- The record team writes the result with no student name.

6 Step 6: Improve control — 4 min

- The fix team changes one moving or soft part.
- The test team repeats the stand test.

7 Step 7: Reset the build — 2 min

- Close the scissors.
- Store the shared prototype as one item.

8 Step 8: Exit check — 3 min

- Name your role contribution.
- Cite one stand test result.

What to turn in

One successful move and how it serves the need.

No computer? Another way to do it

Draw the mechanism and act out the move.

Finished early? Next challenge

Gentle Grip: Modify the gripper to move a paper cup without crushing or dropping it.

Stay safe

Grip only approved lightweight objects and keep the device away from faces, bodies, and breakable items.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.